Type MS · Match human cell structures to their functions

Answer 1 of 10 · Question 7 part a · 6 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

7 (a)

Part	Name of part	Function of part
X	cell membrane;	controls what enters/leaves cell;
Y	nucleus;	controls cell/stores genetic information;
Z	cytoplasm;	site of chemical reactions;

If they get any wrong in the part column, they can score marks for the correct function.

6

(b)

- scrape inside of cheek/cotton bud to remove cheek cells/eq.;
- place (scraping) on slide;
- smear/spread thinly;
- add stain/ methylene blue;
- place coverslip on top;
- remove excess stain;
- sterile/disinfect cotton bud;

4 (

(c)

- ribosomes;
- mitochondria;

accept any suitable structures such as ER, golgi, RER

ignore chromosomes

2

Total = 12

PAPER TOTAL - 90 MARKS

Type MS · Match human cell structures to their functions

Answer 1 of 10 · Question 7 part a · 6 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Match human cell structures to their functions

Answer 2 of 10 · Question 6 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

6(a)	• holds DNA/chromosomes/genetic information in nucleus; • controls what enters and leaves nucleus / allows movement of substances/named substance out of/into the nucleus/exchange of substances between nucleus and cytoplasm;		**(2)**

Question Number	Answer	Additional guidance	Mark
6(b)	B CCG;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(c)	threonine;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(d)	Any six from: • tRNA has anticodons; • translation; • of mRNA; • amino acids carried by tRNA; • joined together; • by peptide bonds; • forming polypeptide/protein; • (final) protein released;	Mp3 must be in context of translation	**(6)**

Question Number	Answer	Additional guidance	Mark
6(e)	• causes change in base sequence/genetic code/named type of mutation; • of DNA;		**(2)**

Total 12 Marks

Type MS · Match human cell structures to their functions

Answer 2 of 10 · Question 6 part a · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Match human cell structures to their functions

Answer 3 of 10 · Question 2 part a.i · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

2 (a)	(i)	<table><tr><th>Part</th><th>Name of part</th><th>Function</th></tr><tr><td>X</td><td>mitochondria;</td><td></td></tr><tr><td>Y</td><td></td><td>controls activities of the cell/stores DNA / genetic information;</td></tr></table>		1 1
	(ii)	**A,** more cell structures can be seen with greater resolution		1
	(iii)	60mm/6cm; 60 ÷ 0.05; 1200;	Ecf two marks for correct calculation from incorrect measurement Full marks for correct final answer	3
(b)		• diffusion; • from a high concentration (in the cell) to a lower concentration (in the blood)/down a concentration gradient;		1 1

Total question 2 = 8 marks

Type MS · Match human cell structures to their functions

Answer 3 of 10 · Question 2 part a.i · 2 marks

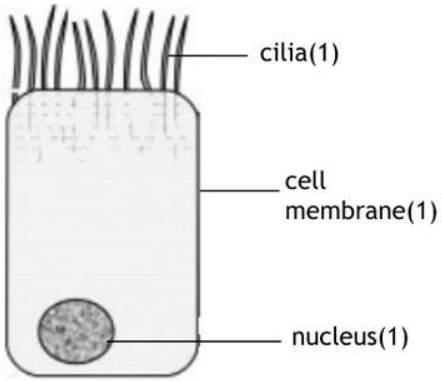
2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match human cell structures to their functions

Answer 4 of 10 · Question 2 part a.ii · 3 marks

2020 NOV · Paper 01 (regional R) · find part a.ii in this mark scheme extract

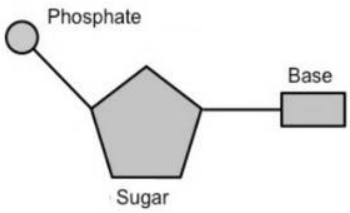
2 (a) (i)			3
(ii)	cilia → holds the genetic code cell membrane → controls entry and exit of nucleus → site where most chemical reactions occur makes proteins moves mucus through the trachea	Reject more than one line from each structure	3
(b) (i)	Tar (1)		1
(ii)	• (cilia)burned/destroyed/reduced in number(1) • paralysed/cannot beat to and fro/move mucus(1)		2

Total for Question 2 = 9 Marks

Type MS · Match human cell structures to their functions

Answer 5 of 10 · Question 8 part a.iii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

8 (a)	(i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
	(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
	(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
	(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
	(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1)  	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)		any three from • deletion(1)	allow description of deletion	3

Type MS · Match human cell structures to their functions

Answer 5 of 10 · Question 8 part a.iii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Match human cell structures to their functions

Answer 6 of 10 · Question 8 part a.ii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
8	(a)	(i)	arrow pointing from top to bottom (1)		1
		(ii)	• provides energy/ATP (1) • for synthesis of acetylcholine/neurotransmitter (1) • movement of vesicles/exocytosis (1)		3
	(b)	(i)	• impulses can't pass to intercostal muscles/diaphragm (1) • unable to contract and move ribs/increase volume of thorax/chest (1)		2
		(ii)	• prevents breakdown of neurotransmitter/acetylcholine (1) • which can compete with curare for/binds to, receptor sites (1)		2

Total for this question = 8 marks

Type MS · Match human cell structures to their functions

Answer 6 of 10 · Question 8 part a.ii · 3 marks

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Type MS · Match human cell structures to their functions

Answer 7 of 10 · Question 2 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
2	(a)	(i)	movement/swimming;	1
		(ii)	C; A, B and D do not release energy	1
		(iii)	three from: • produces a haploid cell / reduces the number of chromosomes by half/to 23; • so at fertilisation the diploid/full number of/46 chromosomes is restored; • prevents doubling of chromosomes at fertilization; • introduces (genetic) variation;	3
	(b)	(i)	sperm egg X Y X XX XY X XX XY ;	One for egg One for sperm One for the offspring 3
		(ii)	Two from: • acrosome; • contains enzymes; • digests/breaks down the egg cell membrane;	2

Total 10 marks

Type MS · Match human cell structures to their functions

Answer 7 of 10 · Question 2 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match human cell structures to their functions

Answer 8 of 10 · Question 1 part a.i · 3 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

Number				
1 (a) (i)	<table><tr><th rowspan="2">cell structure</th><th colspan="3">Function</th></tr><tr><th>makes proteins</th><th>contains DNA</th><th>allows substances in and out of cells</th></tr><tr><td>nucleus</td><td></td><td>✓</td><td></td></tr><tr><td>ribosomes</td><td>✓</td><td></td><td></td></tr><tr><td>cell membrane</td><td></td><td></td><td>✓</td></tr></table>	One mark for each correct row	3
(ii)	cell membrane/cell wall/cytoplasm/plasmid/DNA/ribosomes/flagellum;		1
(b) (i)	Part W phosphate (group); Part X sugar/ deoxyribose sugar;		2
(ii)	• complementary base pairing; • A-T, G-C; • hydrogen bonds between bases;		3
(c)	153 ÷ 3; 51;	Full marks for correct final answer Ecf allow correct final answer to calculation for 1 mark	2

Total for Question 1 = 11 Marks

Type MS · Match human cell structures to their functions

Answer 9 of 10 · Question 4 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Match human cell structures to their functions

Answer 9 of 10 · Question 4 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Match human cell structures to their functions

Answer 10 of 10 · Question 2 part a · 1 mark

2025 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

2	(a)	**C (endoplasmic reticulum)** The answer is not A as the cell membrane transports substance into and out of the cell B cytoplasm as this is a base for chemical reactions D as ribosomes provide sites for protein synthesis		1
	(b) (i)	Z Y W X;		1
	(ii)	growth/repair;	Ignore repair of body cells	1
	(c)	Any two from • mitosis produces diploid cells/46 chromosomes, meiosis produces haploid cells/23 chromosomes; • mitosis produces 2 (daughter) cells, meiosis produces 4 cells/gametes; • (daughter) cells from mitosis are genetically identical, (daughter) cells from meiosis show variation;		2
			Total for Question 2 = 5 marks	